

Photonic THz-ISAC Demonstration with Simultaneous 120Gbit/s Communication and 2.5mm Sensing Resolution

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Abstract We demonstrate a photonic terahertz integrated sensing and communication (THz-ISAC) system, with simultaneously achieving record-high 120 Gbit/s data rate communication and 2.5 mm range sensing resolution at 240-310 GHz band, enabled by the dual-chirp-based integrated waveform and coherent fusion processing.

Introduction

Integrated sensing and communication (ISAC) has been envisioned as a promising technology in the next generation wireless access due to spectral congestion and increasing complexity of existing wireless networks [1]. Compared with traditional all-solid-state electronic systems, the photonics-based approach can provide ultra-broad bandwidth and mitigate in-band harmonic interference [2,3]. Recent photonic ISAC demonstrations were realized with frequency-, time-, and polarization-division multiplexing (FDM, TDM, and PDM). However, they suffer from additional resources overhead, limited time-frequency product, and fail to achieve expected mutual gain [4-11], therefore motivating the research on the integrated waveforms [12-14]. Fig. 1 summarizes the recent demonstrations on photonic ISAC in millimetre-wave (MMW) and THz bands. Amongst those efforts, an electromagnetic PDM-based system has achieved wireless communication of close to 100 Gbit/s, but only with a centimeter-scale resolution [10]. Additionally, enabled by the multiband coherent fusion processing (CFP) algorithm, range resolutions of 9.7 mm and 8.6 mm have been realized, however, only with communication rates below 20 Gbit/s [11,13].

In this paper, we propose and experimentally demonstrate a 3-channel photonic THz-ISAC transmission at 275 GHz with a channel spacing of 25 GHz, each channel combining 16-quadrature amplitude modulation (QAM) or 32-QAM signals with the chirp carrier. The resulting wireless communication rates of up to 120 Gbit/s are higher than other reported photonic ISAC systems. Furthermore, the range sensing resolution record of 2.5 mm is achieved simultaneously using the CFP algorithm. The

proposed photonic THz-ISAC scheme highlights the strong potential of integrated waveforms in high-capacity and high-resolution fibre-wireless transmission systems.

Experimental Setup

The experimental setup for the demonstration of 120 Gbit/s transmission and 2.5 mm range resolution is depicted in Fig. 2(a). A continuous lightwave centered at 1552 nm is radiated from a laser diode (LD), and then launched into an optical phase modulator (PM) to generate the coherent optical frequency comb (OFC). Here, a polarization controller (PC1) is employed to optimize the incident polarization state of the PM. A 25 GHz radio frequency source (RF) is amplified to 34 dBm by a power amplifier (PA) with a 22 dB gain to drive the PM. Subsequently, a programmable wavelength selective switch (WSS) is used to filter out one comb line for the optical local oscillator (LO), and 3 other comb

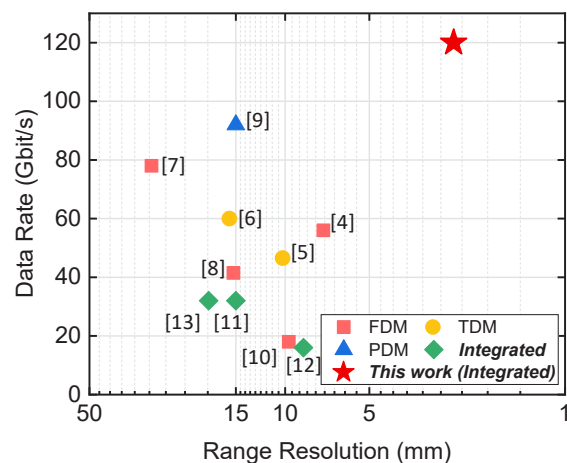


Fig. 1: Recent reported ISAC demonstrations with respect to the data rates and range resolutions.

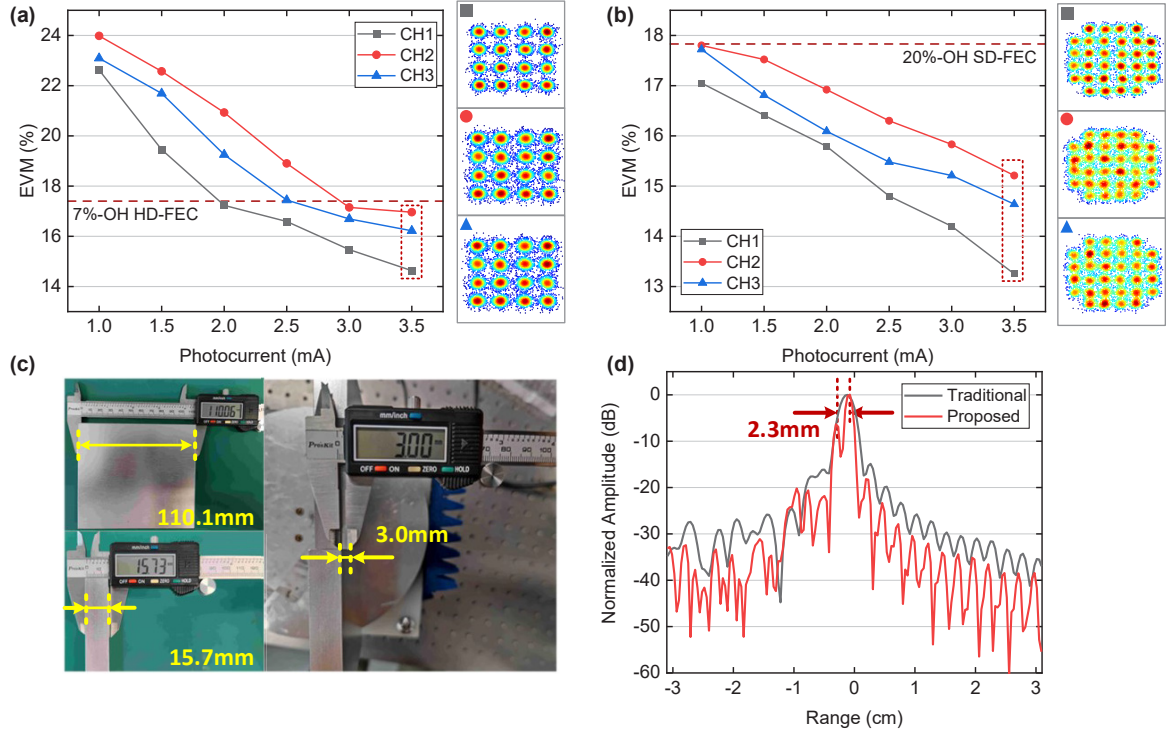


Fig. 3: (a) Measured 3-channel EVM performance of 16-QAM. (b) Measured 3-channel EVM performance of 32-QAM. (c) Photograph of targets. (d) Measured radar range profile for two targets.

error vector magnitude (EVM) performance and the constellations, where 10 GBaud data rate 16-QAM symbols are modulated onto the up-chirp carriers. We can observe that when the photocurrent is greater than 3 mA, all the 3 channels reach the hard-decision forward error correction (HD-FEC) limit with an overhead of 7% [17], resulting in a row data rate of $10 \text{ GBaud} \times 4 \text{ bit/s/Hz} \times 3 \text{ channels} = 120 \text{ Gbit/s}$ with a net data rate of 112.15 Gbit/s. There is a performance penalty between adjacent channels of approximately 0.5 mA, which is caused by the non-flat frequency responses of the optical carriers and the THz devices. Moreover, we also conduct the wireless transmission of 8 GBaud 32-QAM, and the results are shown in Fig. 3(b). As can be seen, the measured EVM can stay below the soft-decision forward error correction (SD-FEC) with 20% overhead in all 3 channels and photocurrents, corresponding to a row data rate of $8 \text{ GBaud} \times 5 \text{ bit/s/Hz} \times 3 \text{ channels} = 120 \text{ Gbit/s}$ with a net data rate of 100 Gbit/s.

The practical sensing performance of the proposed system is also evaluated, as displayed in Fig. 3(c). Here, two static metal targets are placed on a fixed platform, and separated by 3.0 mm in the range direction. Besides, the distance between the transceiver antennas and the platform is set to 0.25 m. Fig. 3(d) shows the measured radar range profile by the traditional approach and the proposed one, respectively. It can be seen that the traditional approach with a

range resolution of 15 mm cannot distinguish the two targets. However, after the CFP processing, there are two clear peaks separated by 0.92 MHz, resulting in a measured distance of $3 \times 10^8 \text{ m/s} \times 0.92 \text{ MHz} \times 1 \mu\text{s} / 60 \text{ GHz} / 2 = 2.3 \text{ mm}$, which is close to the theoretical value of 2.5 mm and exist 0.7 mm error with the measured results in Fig. 3(c), exhibiting the superiority mm-scale range resolution of the proposed system.

Conclusions

In summary, we experimentally demonstrate the wireless transmission of a photonic THz-ISAC system using a dual-chirp-based integrated waveform. A record communication data rate of 120 Gbit/s and a range sensing resolution of 2.5 mm are simultaneously achieved with the integrated waveform, revealing its strong potential in high-capacity and high-resolution fibre-wireless transmission systems.

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